

Exploiting Vulnerabilities in LLMs

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1. Vulnerability Concept & Mechanics

Underlying LLMs can be susceptible to model-level security weaknesses, including prompt injection, jailbreak attempts, training-data memorization, and insecure handling of system instructions. These weaknesses may allow attackers to override intended constraints or cause the model to behave outside its defined operational boundaries.

2. Attack Vector & Exploitation Methodology

System Prompt Extraction

Assess whether adversarial prompting techniques can cause the model to disclose hidden system instructions, configuration details, or security-related guidance.

Jailbreak Testing

Evaluate the model's resistance to adversarial multi-turn prompting, role-play scenarios, encoding techniques, and other attempts to bypass established safety controls.

Instruction Override

Test whether malicious instructions can cause the model to disregard higher-priority instructions or established application constraints.

3. Validation & Impact

- **Impact:** Potential disclosure of proprietary system instructions, safety-control bypass, unintended content generation, and unauthorized model behavior.
- **Validation:** Determine whether security controls can be consistently bypassed and whether the model performs actions or generates content outside its intended policy boundaries.

4. Remediation & Hardening

- Deploy robust LLM safety layers and input-classification mechanisms.
- Separate system instructions from user-controlled input through secure prompt-templating architectures.
- Conduct regular adversarial testing and model evaluations focused on prompt-injection and jailbreak resistance.
- Apply appropriate model alignment and security-tuning techniques to improve robustness against adversarial inputs.